

Reflecting on the Experience at Iowa State University Essay

While working on my COM S 309 project, I have been thinking about a simple way of testing a page before the sign-up and login procedures are completed: a temp button on the main activity. While the entire navigation is being worked out, I could use this button to reach the page and continue work on it; it has become a small part of my project, but it speaks volumes about where I am in my education: I am learning to work around the parts that are not yet completed, or which rely on the work of others

Having transferred to Iowa State University from INTI International College in Malaysia, I have had the opportunity to build on my academic experience both through coursework and through outside opportunities such as a campus IT position, independent cybersecurity studies, and student organization involvement. Each has provided me with ways to engage with the subject matter in different spaces, and also highlight the areas where I need to learn further. As I near the end of my studies, I have a stronger sense of what career path within cybersecurity that I am looking to pursue, and the areas that I need to strengthen for it.

In COM S 309, I am working on the frontend of a team project, some parts of which involve learning the tools while others are focused on ensuring that the team has the information it needs in order to complete its work. We have set up a group chat wherein we can communicate and contacted the teaching assistant about our team size. As development proceeds, it is important to remember what each member is working on, and which parts rely on others' contributions; I am still learning to maintain this communication throughout the duration of a project

By going through Android Studio and Volley tutorials, I have learned to think about the application in more concrete terms. Before I could move forward with developing it further, I had to work through Gradle version issues that prevented it from working correctly. Once I could get the application to run, I was able to experiment with making requests and parsing JSON in order to see how the frontend communicates with the backend; the smaller examples help me to engage with the material at hand, as I am able to tweak one thing and see what changes it concretely. This is what appeals to me about the idea of a temp navigation button: it would give me an opportunity to work with one part of the application while others are being set up.

COM S 363, meanwhile, is asking for a different kind of patience from me: in designing the database, I have to start from a written description and extrapolate the entities and relationships that will be necessary to build out the database. A diagram might look good on paper, but I have to go back to the description and ask myself if what I have designed truly captures what the database needs to do. While experimentation has often been a part of programming, databases ask for a closer reading before actually implementing anything.

In tandem with this, I have continued my independent cybersecurity studies with the Google Cybersecurity Certificate and TryHackMe learning paths, which have allowed me to review concepts at my own pace and engage in practical exercises. When it comes to applying my learning, though, I find that simply finishing a lesson is not always enough of a foundation; I have to be sure that I am able to explain a concept in my own words, and apply it in practice. If I am not, it helps to look at an example that shows the point that I need to review.

As an IT Student Technician with CHHS IT, I have had to balance between being able to understand the technical difficulties that faculty and staff are facing, while also being able to provide them with a suitable explanation and action plan. This ties into my ENGL 302 coursework, in which I have been learning to consider the audience that I am engaging with: when it comes to technical explanations, the audience might not have the same background experience or education in CS or IT, and may need a different depth or style of explanation. What I learn in ENGL 302 is helping me to be more aware of what different audiences might need from me in a technical explanation, and how to provide it.

At the same time, my work in IT highlights some professional concepts that factor into everyday interactions with others: for example, the privileges that one has access to in a system are limited, and it is a professional responsibility to be aware of what one can and cannot do within it. I have to consider my impact within these limitations, and what I am able to confirm or deny based on them. Similarly, if I am unsure of something, it is a responsibility to request assistance. These concepts will be vital to my work in cybersecurity, where considerations of privilege and impact are key parts of designing and maintaining secure systems.

Outside of my technical studies, some of my experiences have raised questions for which a technical approach alone are not enough of an answer. Having studied in both Malaysia and the US, I am often mindful of the diverse backgrounds and expectations of my peers and users. In my entrepreneurship studies, I am being asked to consider factors of cost and value in creating something that fulfills a need. At the same time, my experience with equipment purchasing has taught me to consider the environmental impact of what I am doing: extending the lifespan of equipment wherever possible reduces the cost and e-waste impact of replacing it. I have limited experience considering these factors in conjunction, but I hope to engage further with questions of value and environmental impact in my future work.

With the Hindu YUVA and Indian Student Association, I have been able to learn leadership and communication skills through planning and organizing events, as well as engaging with students from both areas. Event planning has helped me to think through logistics and timelines, as well as adapting to situations that require rearranging plans. Similar considerations come up in my team projects: Someone has to take on the responsibility of relaying updates, clarifying points, and following through on responsibilities in order to keep the team prepared and on schedule. Through peer mentoring, I am able to both practice explaining and engaging with concepts at different levels, while also being able to learn from how other students approach problems.

If I were to go back and speak with my past self, I would have encouraged more note-taking with my experiences. A problem can consume a large amount of my time and energy, only for me to lose the knowledge of what I learned when I move onto the next one. I would have benefited from being able to refer back to what issues I had and what solutions I had tried or found that worked or did not work, and I think it would have helped me to learn to explain what I had done to someone else clearly. For future projects, I would have set aside time to keep up with smaller projects so that I could practice ideas before I actually needed them; it would have helped me to have more time for the material.

For my time at Iowa State, I would like to continue to build my knowledge of networks, Linux, and secure SDLC, and practice whenever I can. I want to document my project work in COM S 309, while also learning from feedback and practicing areas that I need to strengthen. I want to learn more about how professional teams operate in order to troubleshoot and maintain systems as part of an internship, and I am looking forward to continuing to develop my application while I have a frontend to maintain and test alongside my teammates.